

Z_e bypasses compatibility and enters only at reranking



1 Source

VL-SAT / Open3DSG
edge $e = (s, p, o)$
 Z_e : score + rank

2 Evidence

T_e : predicate, classes
 G_e : pair geometry
 Δz , size, offset, gap
 $Z_e \notin C_e$

3 Compatibility

$C_e^{raw} = f_\theta(T_e, G_e)$
explicit $T_e \times G_e$ terms
train-only logistic model
counterfactual controls

4 Reranking

$\tilde{Z}_e$: per source
 $\tilde{C}_e$: per source-family
 $S_{comp} = \tilde{Z}_e \tilde{C}_e$
baseline: $S_{src} = Z_e$

5 Evaluation

3DSSG validation
Recall@K $\uparrow$
Violation@K $\downarrow$
grouped bootstrap CI

Route-specific evidence status

Validated

higher/lower
bigger/smaller

Qualified

left/right
source-dependent

Control + failure

close by; front/behind
support/contact

Not claimed

hidden test; all-family
calibrated abstention